

Brian Maina Kuria

Full-Stack Product Engineer · Mobile · Backend · Payments

[Portfolio](#) | [Github](#) | Nairobi, Kenya | +254 790 408 143 | softdevbriankuria@gmail.com | briankuriamaina@gmail.com

Professional Summary

Full-stack product engineer specializing in mobile and payments. Sole engineer on a live agricultural ERP that disburses real wages over Safaricom Daraja M-Pesa B2C with fully automated reconciliation, and mobile lead on an AI education platform in daily use across Kenyan schools. Four systems shipped to production and running. Strongest in payment integration, offline-first architecture, and building for low-connectivity constraints.

Work Experience

Mobile Lead — CBC AI Platform (EdTech infrastructure, 1,000+ installs, live)

Bombay Softwares, Nairobi · 2023 – Present - [CBC AI Platform](#) · [CBC AI on Play Store](#)

Flutter · Dart · Django REST Framework · PostgreSQL · Bulk SMS · RFID/biometric hardware

- Own the mobile application layer end to end for a production AI platform used daily by Kenyan schools — architecture, design system, state management, API integration and Play Store release engineering — co-developed with a partner engineering team.
- Shipped the school operations core now in daily institutional use: attendance, fee management and financial tracking, timetables, records of work, and teacher–parent SMS communication.
- Drove REST API contracts and performance optimisation across the mobile–backend boundary, and own the release pipeline from development through QA to Google Play.
- Surfaced the platform's intelligence layer on mobile — learner profiles, competency growth, and CBC-aligned assessment reporting — translating national curriculum requirements into shipped product.
- Integrated a hardware data-collection layer (RFID canteen payments, biometric attendance, library RFID) feeding the platform's school knowledge graph.

Sole Engineer — Tea Farm Management System (Agricultural ERP, live production)

Independent engagement, Nairobi · Sept 2025 – Present - [GitHub](#)

Django 5.2 · Python 3.13 · MySQL · Safaricom Daraja M-Pesa B2C · Brevo · cPanel/Phusion Passenger

- Integrated Safaricom Daraja M-Pesa B2C for one-click wage and supplier disbursement, with automated reconciliation between Safaricom transaction IDs and immutable internal audit-ledger records.
- Solved multi-tenant webhook routing for B2C callbacks that carry no application state, by minting dynamic callback URLs with embedded farm IDs at dispatch and matching OriginatorConversationID in idempotent handlers.
- Cut payroll calculation and wage disbursement from 2–3 business days to under 5 minutes by automating the harvest weight → piece-rate wage → mobile payout pipeline, replacing paper ledgers and cash handling.
- Diagnosed and fixed a silent production data-loss bug: Django __date lookups were hitting MySQL CONVERT_TZ() against unpopulated shared-hosting timezone tables, returning NULL and blanking every dashboard. Refactored date-boundary math into timezone-aware Python with exact UTC ranges, making queries host-independent.
- Built 4-tier RBAC with separate portals for SuperAdmin, Farm Owner, Farm Manager and Field Worker, plus a conflict-detecting shift scheduler and dynamic Excel reporting.

ICT Industrial Attachment

Office of the Attorney General & Department of Justice, Nairobi · May – Aug 2026

- Ran the ICT helpdesk for the State Law Office, resolving hardware, software and network requests for staff.
- Configured LAN/internet devices, administered Active Directory accounts on a Windows Server domain, and handled workstation provisioning, backup and recovery, and records digitisation. Written reference available from the Director of ICT.

Projects

Tuko Kadi — Offline-First P2P Multiplayer Card Game · Sole Engineer · v1.0 shipped, August 2026
Flutter 3.x · Dart 3.x · embedded dart:io HttpServer + WebSockets · CustomPainter · AdMob · In-App Purchase

[Tuko Kadi on Play Store](#)

- Engineered zero-infrastructure P2P multiplayer by embedding an HttpServer + WebSocket host on-device; players join by scanning a QR code carrying the host's local IP and room code, reaching sub-5 ms local latency at zero cloud cost with no internet connection.
- Built a deterministic game engine and state machine for cascading card combinations, penalty stacking, Jack deflections, Ace suit calls and Super Ace demands, isolating a pure ValidationEngine behind a negotiation state machine to resolve asynchronous out-of-turn rule interruptions.
- Made networking fault-tolerant with persistent session recovery, automated reconnection, turn-timeout fail-safes, and lobby state-sync that survives players backgrounding their phones.
- Wrote a headless simulator running 500+ full 4-player matches in seconds to verify zero rule violations, deck integrity and bot win-rate balance across four behavioural AI opponents.
- Shipped the full commercial pipeline: AdMob banner/interstitial, IAP premium unlock and tipping tiers, billing error handling, and store release.

Finance Tracker — Multi-Timeframe Financial Dashboard · Sole Engineer · MVP shipped, Nov 2024
Next.js 15 · React 19 · Neon serverless PostgreSQL · Drizzle ORM · Clerk Auth · Recharts

[Finance Tracker Live Site](#)

- Built a multi-timeframe state engine over a normalised 5-table serverless Postgres schema, letting users aggregate weekly, monthly, yearly and custom periods concurrently without stale or hydration-mismatched data.
- Wrote dynamic Drizzle queries using inArray filtering and COALESCE aggregate joins to compute budget-vs-spend across arbitrary period selections at sub-second response times, feeding an 11-scenario heuristic advisory engine.

Education

BSc Software Development — KCA University, Nairobi, Kenya · Expected Nov 2026

Technical Skills

Payments & Integrations — Safaricom Daraja M-Pesa B2C, webhook and callback architecture, idempotent handler design, transaction reconciliation, bulk SMS, Brevo, Clerk Auth, In-App Purchase/Play Billing, AdMob

Mobile — Flutter 3.x, Dart, cross-platform Android & iOS, offline-first architecture, state management, Play Console release management, QR pairing, RFID/biometric hardware

Backend — Python, Django 5, Django REST Framework, FastAPI, Node.js, REST API design, WebSockets, RBAC, Celery-style async patterns

Frontend — TypeScript, JavaScript (ES6+), Next.js 15 (App Router), React 19, Tailwind CSS, Framer Motion, Recharts

Data — PostgreSQL (Neon serverless), MySQL, SQLite, SQL, Django ORM, Drizzle ORM, schema design, aggregation queries

Infrastructure & Practice — Vercel, Railway, cPanel/Phusion Passenger, Gunicorn, WhiteNoise, Git & GitHub, system architecture, state-machine design, automated simulation testing, technical documentation